- Loads from Wind Pressure:

Refer to Regulations and Standards the Saudi Building Code (SBC) 2024

29.3.2.1 Velocity pressure, q_z, evaluated at height z shall be calculated by the following equation:

$$q_z = 0.613\, K_z\, K_{zt}\, K_d\, V^2 \text{ (N/m}^2\text{)} \qquad (29\text{-}1)$$

where,

K_d = Wind directionality factor, see Section 26.6

K_z = Velocity pressure exposure coefficient, see Section 29.3.1

K_zt = Topographic factor, see Section 26.8.2

V = Ultimate wind speed, see Section 26.5.1

q_z = Velocity pressure calculated using Equation (29-1) at height z, (N/m^2)

q_h = Velocity pressure calculated using Equation (29-1) at mean roof height h.

Table 26-1: Wind directionality factor, K_d

Structure Type	Directionality Factor K_d*
Buildings	
Main Wind Force Resisting System	0.85
Components and Cladding	0.85
Arched Roofs	0.85
Chimneys, Tanks, and Similar Structures	
Square	0.90
Hexagonal	0.95
Round	0.95
Solid Freestanding Walls and Solid Freestanding and Attached Signs	0.85
Open Signs and Lattice Framework	0.85
Trussed Towers	
Triangular, square, rectangular	0.85
All other cross sections	0.95

* Directionality Factor K_d has been calibrated with combinations of loads specified in CHAPTER 2. This factor shall only be applied when used in conjunction with load combinations specified in Sections 2.3 and 2.4.

Velocity pressure exposure coefficients, K_h and K_z

Table 29-2

Height above ground level, z (m)	Exposure B	Exposure C	Exposure D
0-5	0.59	0.86	1.04
6	0.62	0.90	1.08
8	0.67	0.95	1.13
10	0.72	1.00	1.18
12	0.76	1.04	1.22
14	0.79	1.07	1.25
16	0.82	1.10	1.28
18	0.85	1.13	1.31
20	0.88	1.16	1.33
22	0.90	1.18	1.35
25	0.93	1.21	1.38
30	0.98	1.26	1.43
35	1.03	1.30	1.47
40	1.07	1.34	1.50
50	1.14	1.40	1.56
60	1.20	1.46	1.61
75	1.28	1.53	1.67
90	1.35	1.59	1.73
105	1.41	1.64	1.77
120	1.46	1.69	1.82
135	1.51	1.73	1.85
150	1.56	1.77	1.89

26.8.2 Topographic Factor.

26.8.2.1 The wind speed-up effect shall be included in the calculation of design wind loads by using the factor K_{zt}

$$K_{zt} = (1 + K_1 K_2 K_3)^2 \qquad (26\text{-}1)$$

where K_1, K_2, and K_3 are given in Figure 26-3.

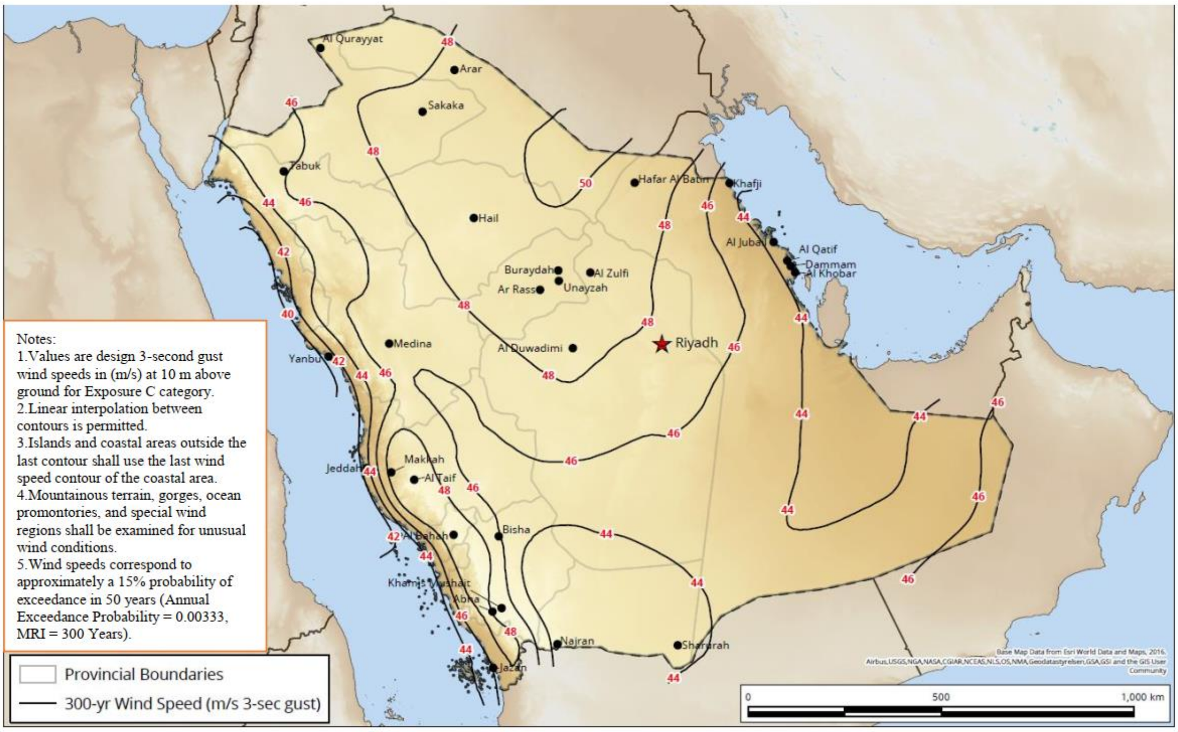

Figure 26-2C: Ultimate wind speeds for Risk Category I Buildings and other Structures.

26.8.2.2 If site conditions and locations of buildings and other structures do not meet all the conditions specified in Section 26.8.1 then $K_{zt} = 1.0$.

29.5.1 The design wind force for other structures (chimneys, tanks, rooftop equipment for $h > 18\,m$, and similar structures, open signs, lattice frameworks, and trussed towers) shall be determined by the following equation:

$$F = q_z\, G\, C_f\, A_f \quad (\mathrm{N}) \qquad (29\text{-}3)$$

where,

q_z = Velocity pressure evaluated at height z as defined in Section 29.3, of the centroid of area A_f

G = Gust-effect factor from Section 26.9

C_f = Force from Figure 29-2 through Figure 29-4

A_f = Projected area normal to the wind except where C_f is specified for the actual surface area, in m²

26.9 —Gust-effects

26.9.1 Gust-Effect Factor. The gust-effect factor for a rigid building or other structure is permitted to be taken as 0.85.

Design wind loads—Other structures		All heights
Figure 29-2	**Force coefficients, C_f**	**Chimneys, tanks, rooftop Equipment, & similar structures**

Cross-Section	Type of Surface	h/D 1	h/D 7	h/D 25
Square (wind normal to face)	All	1.3	1.4	2.0
Square (wind along diagonal)	All	1.0	1.1	1.5
Hexagonal or octagonal	All	1.0	1.2	1.4
Round ($D\sqrt{q_z} > 5.3$, D in m, q_z in N/m²)	Moderately smooth	0.5	0.6	0.7
	Rough (D'/D = 0.02)	0.7	0.8	0.9
	Very rough (D'/D = 0.08)	0.8	1.0	1.2
Round ($D\sqrt{q_z} \le 5.3$, D in m, q_z in N/m²)	All	0.7	0.8	1.2

Notes:

1. The design wind force shall be calculated based on the area of the structure projected on a plane normal to the wind direction. The force shall be assumed to act parallel to the wind direction.
2. Linear interpolation is permitted for h/D values other than shown.
3. Notation:
   - D: diameter of circular cross-section and least horizontal dimension of square, hexagonal or octagonal cross-sections at elevation under consideration, in meters;
   - D': depth of protruding elements such as ribs and spoilers, in meters; and
   - h: height of structure, in meters; and
   - q_z: velocity pressure evaluated at height z above ground, in N/m².
4. For rooftop equipment on buildings with a mean roof height of $h \le 18$ m., use Section 29.5.2.

Table 1-2: Risk category of buildings and other structures for flood, wind and earthquake loads

RISK CATEGORY	NATURE OF OCCUPANCY
I	Buildings and other structures that represent a low hazard to human life in the event of failure, including but not limited to: • Agricultural facilities. • Certain temporary facilities. • Minor storage facilities.
II	Buildings and other structures except those listed in Risk Categories I, III and IV.
III	Buildings and other structures that represent a substantial hazard to human life in the event of failure, including but not limited to: • Buildings and other structures whose primary occupancy is public assembly with an occupant load greater than 300. • Buildings and other structures containing Group E occupancies with an occupant load greater than 250. • Buildings and other structures containing educational occupancies for students above the 12^{th} grade with an occupant load greater than 500. • Group I-2 occupancies with an occupant load of 50 or more resident care recipients but not having surgery or emergency treatment facilities. • Group I-3 occupancies. • Any other occupancy with an occupant load greater than 5,000.[a] • Power-generating stations, water treatment facilities for potable water, wastewater treatment facilities and other public utility facilities not included in Risk Category IV. • Buildings and other structures not included in Risk Category IV containing quantities of toxic or explosive materials that: Exceed maximum allowable quantities per control area as given in Table 307.1(1) or 307.1(2) of SBC 201 or per outdoor control area in accordance with the *SBC 801*; and Are sufficient to pose a threat to the public if released.[b]
IV	Buildings and other structures designated as essential facilities, including but not limited to: • Group I-2 occupancies having surgery or emergency treatment facilities. • Fire, rescue, ambulance and police stations and emergency vehicle garages. • Designated earthquake, hurricane or other emergency shelters. • Designated emergency preparedness, communications and operations centers and other facilities required for emergency response. • Power-generating stations and other public utility facilities required as emergency backup facilities for Risk Category IV structures. • Buildings and other structures containing quantities of highly toxic materials that: Exceed maximum allowable quantities per control area as given in Table 307.1(2) of SBC 201 or per outdoor control area in accordance with the *SBC 801*; and Are sufficient to pose a threat to the public if released.[b] • Aviation control towers, air traffic control centers and emergency aircraft hangars. • Buildings and other structures having critical national defense functions. • Water storage facilities and pump structures required to maintain water pressure for fire suppression.

a. For purposes of occupant load calculation, occupancies required by Table 1004.1.2 of SBC 201 to use gross floor area calculations shall be permitted to use net floor areas to determine the total occupant load.

b. Where approved by the building official, the classification of buildings and other structures as Risk Category III or IV based on their quantities of toxic, highly toxic or explosive materials is permitted to be reduced to Risk Category II, provided it can be demonstrated by a hazard assessment in accordance with Section 1.6.3 that a release of the toxic, highly toxic or explosive materials is not sufficient to pose a threat to the public.